

Alexei Oblomkov

RESEARCH INTERESTS

Representation theory, algebraic geometry and mathematical physics.

EDUCATION

2005	Massachusetts Institute of Technology	Ph.D. in mathematics Advisor: Pavel Etingof
1999-2001	Independent University of Moscow	Graduate Study
1999-2001	Moscow State University	Graduate Study
1998	Independent University of Moscow	MA in Pure Mathematics
1998	Moscow State University	MA in Pure Mathematics

EMPLOYMENT

2018 Spring	MSRI	Research Professor
2015-	University of Massachusetts, Amherst	Associate Professor
2009-2015	University of Massachusetts, Amherst	Assistant Professor
2008-2009	Princeton University	Visiting Assistant Professor
2006-2008	Princeton University	Instructor
2005-2006	Institute for Advanced Study	Postdoctoral Fellow

GRANTS

2018-2021	NSF FRG DMS-1760373,	\$300,000, for UMass
2015-2016	NSF DMS-1546311, Workshop funding	\$25,000, sole co-PI
2014-2019	NSF CAREER DMS-1352398	\$420,000.00, sole PI
2010-2013	NSF DMS-1001609	\$123,910.00, sole PI
2007-2010	NSF DMS-0701367	\$111,300.00, sole PI

AWARDS

2018	Simons Fellow in Mathematics for Fall 2018
2014-2019	NSF CAREER Award
2010-2012	Sloan Fellowship
2004	Charles and Jennifer Johnson Prize, MIT departmental award for the best paper submitted by a graduate student
2004, 2003	Rogers Family Prize, MIT departmental award for the best mentor-mentee team in Summer Program for Undergraduate research

PH.D. ADVISING

1. My student Tom Shelly graduated in August 2016 and started his job as Visiting Assistant Professor at Mount Holyoke College.
2. My student Toby Willson graduated in December 2015 and started his job Bloomberg.
3. I currently work with Arthur Wang.

SELECTED PUBLICATIONS

1. (with J. V. Stokman) *Vector valued spherical function and Macdonald-Koornwinder polynomials*, math.QA/0311512, Compositio Math. 141, (2005), no. 5, 1310–1350.
2. (with P. Etingof, E. Rains) *Generalized double affine Hecke algebras of rank 1 and quantized Del Pezzo surfaces*, math.QA/0406480, Advances in Mathematics, 212 (2007), no. 10, 749–796.
3. (with P. Etingof, W. L. Gan, V. Ginzburg) *Harish-Chandra homomorphisms and symplectic reflection algebras for wreath-products*, math.RT/0511489, Publications Mathématiques de l’IHÉS, No. 105 (2007), 91–155.
4. (with D. Maulik) *Quantum cohomology of $\text{Hilb}_m(A_n)$* , arXiv:0802.2737, Journal of AMS, vol. 22 (2009), 1055–1091.
5. (with D. Maulik, A. Okounkov, R. Pandharipande) *The Gromov-Witten/Donaldson-Thomas correspondence for toric 3-folds*, arXiv:0809.3976, Invent. Math. 186 (2011), no. 2, 434–479.
6. (with V. Shende) *The Hilbert scheme of a plane curve singularity and the HOMFLY polynomial of its link*, arxiv:1003.1568, Duke Math. J. 161 (2012), no. 7, 1271–1303.
7. (with E. Gorsky, V. Shende, J. Rassmusen), *Torus knots and Rational DAHA*, arXiv:1207.4523, Duke Math. J. 163, (2014), no. 15, 325–401.
8. (with Z. Yun) *Geometric Representations of graded and rational Cherednik algebras*, arXiv:1407.5685, Adv. Math. 292 (2016), 601–706.
9. (with L. Rozansky) *Knot Homology and sheaves on the Hilbert scheme of points on the plane*, Selecta Math. (N.S.) 24 (2018), no. 3, 2351–2454, arXiv:1608.03227.
10. (with A. Okounkov, R. Pandharipande) *GW/PT descendent correspondence via vertex operators*, accepted in Communications in Mathematical Physics, arXiv:1806.00714.

PROFESSIONAL SERVICE

Refereeing: Communications in Mathematical Physics, Contemporary Mathematics, Transformation Groups, IMRN, JAMS, Annals of Mathematics, Journal of Algebraic Combinatorics A, Advances in Mathematics, Transactions of AMS, Compositio, Journal of Algebra, Selecta, Inventiones.

Refereeing Grant Proposals: NSA, Swiss National Science Foundation, the Netherlands Organisation for Scientific Research, the Chilean National Science and Technology Commission, Newton Institute Scientific proposals, served on NSF panel.

Conference organization: PI co-organized one-week workshop in Banff (June 2014), co-organized three-day workshop in Amherst (October 2015), co-organized three-day workshop in Amherst (July 2017), co-organize one-week AIM workshop (October 2018), co-organized one-week FRG workshop at UC Davis (June 2019).